

Lipkin Hamiltonian to Pauli Strings

Quasispin Representation, Binary Encoding, and Direct Qubit Mapping

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We want to rewrite the Lipkin Hamiltonian

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + \frac{V}{2} (J_+^2 + J_-^2) + \frac{W}{2} J_z^2$$

in terms of tensor products of Pauli operators I, X, Y, Z .

Two cases will be treated:

- 1 $N = 4$ particles in the quasispin $j = 2$ sector, encoded on **3 qubits** (compressed binary encoding).
- 2 $N = 6$ particles represented on **6 qubits** (direct collective-spin realization).

The emphasis is on *how* the mapping is constructed, not only on the final formulas.

The quasispin algebra

The Lipkin model is naturally written in terms of collective quasispin operators

$$J_x, \quad J_y, \quad J_z,$$

with ladder operators

$$J_{\pm} = J_x \pm iJ_y.$$

They satisfy the $su(2)$ commutation relations

$$[J_x, J_y] = iJ_z, \quad [J_y, J_z] = iJ_x, \quad [J_z, J_x] = iJ_y,$$

or equivalently

$$[J_z, J_{\pm}] = \pm J_{\pm}, \quad [J_+, J_-] = 2J_z.$$

Useful algebraic simplification

A key identity is

$$J_+^2 + J_-^2 = (J_x + iJ_y)^2 + (J_x - iJ_y)^2 = 2(J_x^2 - J_y^2).$$

Therefore the Hamiltonian may also be written as

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + V (J_x^2 - J_y^2) + \frac{W}{2} J_z^2.$$

This form is particularly convenient for direct qubit mappings.

Quasispin basis

In the fixed- j representation, the natural basis is

$$|j, m\rangle, \quad m = -j, -j+1, \dots, j.$$

The standard actions are

$$\begin{aligned} J_z |j, m\rangle &= m |j, m\rangle, \\ J_+ |j, m\rangle &= \sqrt{j(j+1) - m(m+1)} |j, m+1\rangle, \\ J_- |j, m\rangle &= \sqrt{j(j+1) - m(m-1)} |j, m-1\rangle. \end{aligned}$$

Thus J_z is diagonal, while $J_{\pm}$ shift m by one unit.

Strategy for producing Pauli strings

There are two standard routes:

- 1 **Compressed encoding:** encode the quasispin basis $|j, m\rangle$ directly into computational basis states of a smaller qubit register.
- 2 **Direct qubit mapping:** express the collective operators as sums over physical qubit Pauli operators and expand the Hamiltonian.

In these notes:

- the $N = 4$ case uses route 1,
- the $N = 6$ case uses route 2.

Case $N = 4$: quasispin dimension

For $N = 4$, the quasispin is

$$j = \frac{N}{2} = 2.$$

Hence the Hilbert space dimension in the $j = 2$ sector is

$$2j + 1 = 5.$$

Since

$$2^2 = 4 < 5 \leq 8 = 2^3,$$

the smallest binary encoding requires **3 qubits**.

Choice of binary encoding for $N = 4$

We choose the map

$$|j = 2, m = -2\rangle \leftrightarrow |000\rangle ,$$

$$|j = 2, m = -1\rangle \leftrightarrow |001\rangle ,$$

$$|j = 2, m = 0\rangle \leftrightarrow |010\rangle ,$$

$$|j = 2, m = 1\rangle \leftrightarrow |011\rangle ,$$

$$|j = 2, m = 2\rangle \leftrightarrow |100\rangle .$$

The remaining computational states

$$|101\rangle , |110\rangle , |111\rangle$$

are unphysical and should be excluded or penalized.

Projectors in terms of Pauli Z

For one qubit,

$$|0\rangle\langle 0| = \frac{I + Z}{2}, \quad |1\rangle\langle 1| = \frac{I - Z}{2}.$$

Thus for a three-qubit computational basis state $|abc\rangle$,

$$|abc\rangle\langle abc| = \frac{1}{8}(I + (-1)^a Z_2)(I + (-1)^b Z_1)(I + (-1)^c Z_0).$$

This gives a systematic way to build diagonal operators such as J_z and J_z^2 .

Transition operators in terms of X and Y

For one qubit,

$$|1\rangle\langle 0| = \frac{X - iY}{2}, \quad |0\rangle\langle 1| = \frac{X + iY}{2}.$$

Therefore an off-diagonal transition such as

$$|a_2 a_1 a_0\rangle\langle b_2 b_1 b_0|$$

is written as a tensor product of:

- projectors $\frac{I \pm Z}{2}$ on unchanged bits,
- ladder pieces $\frac{X \mp iY}{2}$ on flipped bits.

This is the basic rule for expanding J_+ , J_- , and $J_+^2 + J_-^2$.

Matrix form of J_z for $j = 2$

In the encoded basis, J_z is diagonal:

$$J_z = \sum_{m=-2}^2 m |m\rangle \langle m|.$$

With the chosen binary map, this becomes

$$J_z = -2 |000\rangle \langle 000| - |001\rangle \langle 001| + 0 |010\rangle \langle 010| + |011\rangle \langle 011| + 2 |100\rangle \langle 100|.$$

We take the operator to vanish on the three unused basis states.

Explicit Pauli decomposition of J_z (3 qubits)

Expanding the projectors gives

$$J_z = -\frac{1}{4}IZI + \frac{1}{4}IZZ - \frac{1}{2}ZII - \frac{1}{2}ZIZ - \frac{3}{4}ZZI - \frac{1}{4}ZZZ.$$

Here the notation means, for example,

$$IZZ = I \otimes Z \otimes Z, \quad ZII = Z \otimes I \otimes I.$$

This formula is exact on the chosen 5-dimensional encoded subspace.

Explicit Pauli decomposition of J_z^2 (3 qubits)

Since

$$J_z^2 = \sum_{m=-2}^2 m^2 |m\rangle \langle m|,$$

we obtain

$$J_z^2 = \frac{5}{4} III + \frac{3}{4} IIZ + IZI + IZZ + \frac{1}{4} ZII - \frac{1}{4} ZIZ.$$

This again acts exactly on the chosen encoded subspace and is set to zero on unused basis states.

Action of the ladder operators for $j = 2$

For $j = 2$,

$$\begin{aligned} J_+ |-2\rangle &= 2 |-1\rangle, & J_+ |-1\rangle &= \sqrt{6} |0\rangle, \\ J_+ |0\rangle &= \sqrt{6} |1\rangle, & J_+ |1\rangle &= 2 |2\rangle, & J_+ |2\rangle &= 0. \end{aligned}$$

Thus

$$J_+ = 2 |001\rangle \langle 000| + \sqrt{6} |010\rangle \langle 001| + \sqrt{6} |011\rangle \langle 010| + 2 |100\rangle \langle 011|.$$

And $J_- = J_+^\dagger$.

Why $J_+^2 + J_-^2$ is simpler than J_+ itself

Because J_+^2 changes m by two units, it connects only

$$|-2\rangle \leftrightarrow |0\rangle, \quad |-1\rangle \leftrightarrow |1\rangle, \quad |0\rangle \leftrightarrow |2\rangle.$$

For $j = 2$,

$$J_+^2 |-2\rangle = 2\sqrt{6} |0\rangle,$$

$$J_+^2 |-1\rangle = 6 |1\rangle,$$

$$J_+^2 |0\rangle = 2\sqrt{6} |2\rangle.$$

Therefore

$$J_+^2 + J_-^2$$

is real and Hermitian, and its Pauli decomposition contains only real coefficients.

Explicit $J_+^2 + J_-^2$ decomposition for $N = 4$

The exact three-qubit decomposition is

$$\begin{aligned} J_+^2 + J_-^2 = & \frac{3 + \sqrt{6}}{2} IXI + \frac{\sqrt{6} - 3}{2} IXZ + \frac{\sqrt{6}}{2} XXI + \frac{\sqrt{6}}{2} XXZ \\ & + \frac{\sqrt{6}}{2} YYI + \frac{\sqrt{6}}{2} YYZ + \frac{3 + \sqrt{6}}{2} ZXI + \frac{\sqrt{6} - 3}{2} ZXZ. \end{aligned}$$

This is the key off-diagonal part of the Lipkin Hamiltonian in the compressed 3-qubit encoding.

Final $N = 4$, 3-qubit Hamiltonian

Using

$$H = \frac{\epsilon}{2}(J_z - 2) + \frac{V}{2}(J_+^2 + J_-^2) + \frac{W}{2}J_z^2,$$

the encoded Hamiltonian is

$$\begin{aligned} H_{N=4}^{(3q)} = & \left(-\epsilon + \frac{5W}{8}\right) III + \frac{3W}{8} IIZ + \left(-\frac{\epsilon}{8} + \frac{W}{2}\right) IZI + \left(\frac{\epsilon}{8} + \frac{W}{2}\right) IZZ \\ & + \left(-\frac{\epsilon}{4} + \frac{W}{8}\right) ZII + \left(-\frac{\epsilon}{4} - \frac{W}{8}\right) ZIZ - \frac{3\epsilon}{8} ZZI - \frac{\epsilon}{8} ZZZ \\ & + \frac{V(3 + \sqrt{6})}{4} IXI + \frac{V(\sqrt{6} - 3)}{4} IXZ + \frac{V\sqrt{6}}{4} XXI + \frac{V\sqrt{6}}{4} XXZ \\ & + \frac{V\sqrt{6}}{4} YYI + \frac{V\sqrt{6}}{4} YYZ + \frac{V(3 + \sqrt{6})}{4} ZXI + \frac{V(\sqrt{6} - 3)}{4} ZXZ. \end{aligned}$$

Physical-subspace penalty for the 3-qubit encoding

Because only five of the eight computational basis states are physical, it is often useful to add a penalty

$$H_{\text{pen}} = \lambda P_{\text{unphys}},$$

where

$$P_{\text{unphys}} = |101\rangle\langle 101| + |110\rangle\langle 110| + |111\rangle\langle 111|.$$

This suppresses leakage into unused basis states. Each projector can again be written in terms of I and Z only.

Interpretation of the 3-qubit mapping

The compressed mapping is efficient because:

- the $j = 2$ quasispin sector has dimension only 5,
- binary encoding uses only $\lceil \log_2 5 \rceil = 3$ qubits,
- all quasispin operators become finite sums of Pauli strings.

The price is that one must carefully enforce the physical code space.

Case $N = 6$: direct collective-spin realization

For $N = 6$, one natural choice is to use one qubit per particle. Then the collective quasispin operators are represented directly as

$$J_x = \frac{1}{2} \sum_{i=1}^6 X_i, \quad J_y = \frac{1}{2} \sum_{i=1}^6 Y_i, \quad J_z = \frac{1}{2} \sum_{i=1}^6 Z_i,$$

where X_i, Y_i, Z_i act on qubit i .

Collective spin squares

Because $X_i^2 = Y_i^2 = Z_i^2 = I$,

$$J_x^2 = \frac{1}{4} \sum_{i=1}^6 I + \frac{1}{4} \sum_{i \neq j} X_i X_j = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} X_i X_j.$$

Similarly,

$$J_y^2 = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} Y_i Y_j,$$

$$J_z^2 = \frac{6}{4} I + \frac{1}{2} \sum_{i < j} Z_i Z_j.$$

Rewrite the interaction term

Using

$$J_+^2 + J_-^2 = 2(J_x^2 - J_y^2),$$

we get

$$\frac{V}{2}(J_+^2 + J_-^2) = V(J_x^2 - J_y^2).$$

Substituting the previous expansions,

$$J_x^2 - J_y^2 = \frac{1}{2} \sum_{i < j} (X_i X_j - Y_i Y_j),$$

since the identity contributions cancel. Therefore

$$\frac{V}{2}(J_+^2 + J_-^2) = \frac{V}{2} \sum_{i < j} (X_i X_j - Y_i Y_j).$$

Rewrite the linear J_z term

We have

$$\frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) = \frac{\epsilon}{2} \left(\frac{1}{2} \sum_{i=1}^6 Z_i - 3I \right).$$

Hence

$$\frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) = \frac{\epsilon}{4} \sum_{i=1}^6 Z_i - \frac{3\epsilon}{2} I.$$

Rewrite the J_z^2 term

Using

$$J_z^2 = \frac{6}{4}I + \frac{1}{2} \sum_{i < j} Z_i Z_j,$$

we obtain

$$\frac{W}{2} J_z^2 = \frac{3W}{4} I + \frac{W}{4} \sum_{i < j} Z_i Z_j.$$

Final $N = 6$, 6-qubit Hamiltonian

Collecting all pieces,

$$H_{N=6}^{(6q)} = \left(-\frac{3\epsilon}{2} + \frac{3W}{4} \right) I + \frac{\epsilon}{4} \sum_{i=1}^6 Z_i + \frac{V}{2} \sum_{i < j} (X_i X_j - Y_i Y_j) + \frac{W}{4} \sum_{i < j} Z_i Z_j.$$

This is the clean direct Pauli-string representation on 6 qubits.

How many Pauli terms appear for $N = 6$?

For 6 qubits:

- one identity term,
- 6 one-body Z_i terms,
- $\binom{6}{2} = 15$ two-body $X_i X_j$ terms,
- 15 two-body $Y_i Y_j$ terms,
- 15 two-body $Z_i Z_j$ terms.

So the total number of Pauli strings is

$$1 + 6 + 15 + 15 + 15 = 52.$$

This is still highly structured, since all coefficients come from only three parameters ϵ, V, W .

Explicit pairwise form for the interaction term

The interaction contribution is

$$\frac{V}{2} (X_1 X_2 - X_1 Y_2)$$

No—that is not the correct structure.

The correct two-body interaction is

$$\frac{V}{2} \sum_{1 \leq i < j \leq 6} (X_i X_j - Y_i Y_j).$$

Expanded explicitly, it includes pairs such as

$$X_1 X_2 - Y_1 Y_2, \quad X_1 X_3 - Y_1 Y_3, \quad \dots, \quad X_5 X_6 - Y_5 Y_6.$$

This pairwise all-to-all form is a hallmark of the direct collective-spin mapping.

Direct mapping versus compressed mapping

For $N = 6$, the direct 6-qubit representation has advantages:

- extremely simple operator formulas,
- immediate interpretation in terms of collective spin,
- no leakage into unphysical binary code states.

Its disadvantage is qubit cost:

- the physical symmetric subspace has dimension only $N + 1 = 7$,
- but the full register has dimension $2^6 = 64$.

So this is simple but not qubit-optimal.

Comparison of the two mappings

Case	Qubits used	Main advantage
$N = 4$, compressed	3	minimal qubit count
$N = 6$, direct	6	simplest Pauli expansion

More conceptually:

- compressed encodings reduce qubit count but require code-space management,
- direct collective encodings use more qubits but produce transparent many-body Pauli Hamiltonians.

General recipe for quasispin-to-Pauli conversion

To convert a Lipkin quasispin Hamiltonian to Pauli strings:

- 1 Choose an encoding of the $|j, m\rangle$ basis into computational basis states.
- 2 Express diagonal operators (J_z, J_z^2) using computational basis projectors.
- 3 Express off-diagonal operators ($J_{\pm}, J_{\pm}^2$) using transition operators built from X and Y .
- 4 Simplify using

$$J_+^2 + J_-^2 = 2(J_x^2 - J_y^2)$$

whenever a direct collective-spin realization is available.

- 5 Collect coefficients into sums of Pauli strings.

We have shown how to transform the Lipkin Hamiltonian

$$H = \frac{\epsilon}{2} \left(J_z - \frac{N}{2} \right) + \frac{V}{2} (J_+^2 + J_-^2) + \frac{W}{2} J_z^2$$

into Pauli-string form in two concrete cases:

- $N = 4$: exact **3-qubit compressed binary encoding** of the $j = 2$ quasispin sector,
- $N = 6$: exact **6-qubit direct collective-spin mapping**.

The central ideas are:

- diagonal operators from Z -projectors,
- transitions from X and Y ,
- collective-spin identities for simplification.

End

Questions?